THE LOWER SHRIEK COMPUTED BY A POLYNOMIAL SIMPLICIAL RESOLUTION WITH TRIVIAL KAN FIBRATION AUGMENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries two displays (the isomorphism and the reference to the functoriality in F); the proof works inside the category C fixed earlier in the chapter.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The cosimplicial resolution criterion for computing lower shriek. Cohomology on Sites, Lemma sites-cohomology-lemma-compute-by-cosimplicial-resolution, used at proof lines 458-465. Reduces the claim to the homotopy statement about S_bullet.
- Stability of trivial Kan fibrations under base change and finite products. Simplicial, Lemmas simplicial-lemma-trivial-kan-base-change and simplicial-lemma-product-trivial-kan, used at proof lines 481-485. Give that the fibres and their product over E are trivial Kan fibrations.
- A trivial Kan fibration is homotopy equivalent to a point. Simplicial, Lemma simplicial-lemma-trivial-kan-homotopy, used at proof lines 486-487. Concludes the homotopy statement.
- Every object of the category C is a polynomial algebra over A. used at proof lines 466-467. Allows writing Q = A[E].

Source reference. Original lemma 08NR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10999–10999. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08NN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10996–10996. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08NS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11000–11000. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08Q9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11052–11052. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *With notation as above let $P_\bullet$ be a simplicial A-algebra endowed with an augmentation $\epsilon : P_\bullet \to B$. Assume each P_n is a polynomial algebra over A and ϵ is a trivial Kan fibration on underlying simplicial sets. Then*

$$L\pi_!(\mathcal{F}) = \mathcal{F}(P_\bullet, \epsilon)$$

in $D(Ab)$, resp. $D(B)$ functorially in $\mathcal{F}$ in $Ab(\mathcal{C})$, resp. $Mod(\underline{B})$.

Proof. We will use the criterion of Cohomology on Sites, Lemma 08Q9 to prove this. Given an object $U = (Q, \beta)$ of $\mathcal{C}$ we have to show that

$$S_\bullet = \mathrm{Mor}_{\mathcal{C}}((P_\bullet, \epsilon), (Q, \beta))$$

is homotopy equivalent to a singleton. Write $Q = A[E]$ for some set E (this is possible by our choice of the category $\mathcal{C}$). We see that

$$S_\bullet = \mathrm{Mor}_{Sets}((E, \beta|_E), (P_\bullet, \epsilon))$$

Let $*$ be the constant simplicial set on a singleton. For $b \in B$ let $F_{b,\bullet}$ be the simplicial set defined by the cartesian diagram

$$\begin{array}{ccc} F_{b,\bullet} & \longrightarrow & P_\bullet \\ \downarrow & & \downarrow{\scriptstyle \epsilon} \\ * & \xrightarrow{b} & B \end{array}$$

With this notation $S_\bullet = \prod_{e \in E} F_{\beta(e),\bullet}$. Since we assumed ϵ is a trivial Kan fibration we see that $F_{b,\bullet} \to *$ is a trivial Kan fibration (Simplicial, Lemma 08NN). Thus $S_\bullet \to *$ is a trivial Kan fibration (Simplicial, Lemma 08NR). Therefore $S_\bullet$ is homotopy equivalent to $*$ (Simplicial, Lemma 08NS). □